RAFFLES INSTITUTION
H2 Mathematics (9758)
2026 Year 6

Chapter 9C: Differential Equations

SYLLABUS INCLUDES

- Solving for the general solutions and particular solutions of differential equations of the form $\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{f}(x)\mathrm{g}(y)$, including reducing a given differential equation to this form by means of a given substitution
- Formulating a differential equation from a problem situation
- Interpreting a differential equation and its solution in terms of a problem situation

PRE-REQUISITES

- Differentiation and Integration

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1 Introduction

Many of the fundamental laws of physics, chemistry, biology and economics can be formulated as differential equations. These equations express the relationship involving the rates of change of continuously changing quantities modeled by functions and are used whenever a rate of change (derivative) is known but the originating process is not. One of the key applications of differential equations is in modeling of real-life phenomenon, where researchers seek to find and understand the relationships between variables and to predict future behaviour. For example, suppose an experimenter observes that in a particular interval of time,

> "The **rate of increase** of the population of a certain insect is **proportional** to the **number** of insects (x) alive at any time t."

Formulating this mathematically as a **differential equation** in x:

> If x is the number of insects alive at any time t,
>
> then $\frac{\mathrm{d}x}{\mathrm{d}t} \propto x$ or $\frac{\mathrm{d}x}{\mathrm{d}t} = kx$ where k is a positive constant.

To predict or project future population numbers using this model, one would need a way to express x as a function of t.

1.1 Definitions

Let y be a function of x. Then any equation relating x, y and one or more of the derivatives $\frac{\mathrm{d}y}{\mathrm{d}x}$, $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}, \ldots, \frac{\mathrm{d}^ny}{\mathrm{d}x^n}, n \in \mathbb{Z}^+$ is called a **Differential Equation (DE)**.

Some examples of Differential Equations:

(1) $\frac{\mathrm{d}y}{\mathrm{d}x} = 2$ (2) $x\frac{\mathrm{d}y}{\mathrm{d}x} + y = 0$ (3) $(\sec x)\frac{\mathrm{d}y}{\mathrm{d}x} + y^2 = 0$

(4) $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 3\frac{\mathrm{d}y}{\mathrm{d}x} = y$ (5) $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^3 = xy$ (6) $\left(\frac{\mathrm{d}^3y}{\mathrm{d}x^3}\right)^2 = \left(\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right)^3$

Order of a Differential Equation

The **order** of a differential equation is the highest derivative present in the equation. In the above examples, (1), (2) and (3) are first order DEs, (4) and (5) are second order DEs, and (6) is a third order DE. The DEs discussed in this chapter are first order DEs.

Linear Differential Equation

A differential equation is said to be linear if it can be expressed in the form:

$$a_n(x)\frac{\mathrm{d}^n y}{\mathrm{d}x^n} + a_{n-1}(x)\frac{\mathrm{d}^{n-1} y}{\mathrm{d}x^{n-1}} + \ldots + a_1(x)\frac{\mathrm{d}y}{\mathrm{d}x} + a_0(x)y + b(x) = 0,$$

where $a_n(x)$, $a_{n-1}(x)$, …, $a_1(x)$, $a_0(x)$ and $b(x)$ are functions in terms of x (not y).

1.2 General Solutions of Differential Equations

We say that $y = \mathrm{f}(x)$ is a **solution** of a differential equation if f and its derivatives satisfy the differential equation.

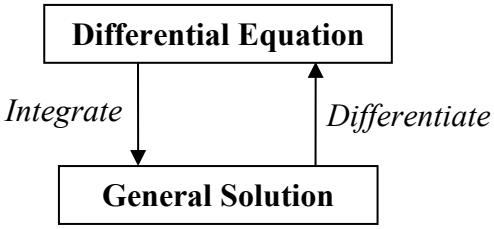
Given that $\frac{\mathrm{d}y}{\mathrm{d}x} = 2x$, write down a solution that solves the differential equation.

$y = x^2 + 1$

One will realize that any equation of the form $y = x^2 + c$ satisfies the above differential equation, where c is an arbitrary real constant. Hence $y = x^2 + c$ is said to be the **general solution** to the differential equation $\frac{\mathrm{d}y}{\mathrm{d}x} = 2x$, as it encompasses *all* possible functions that satisfy the differential equation.

The **general solution** of a differential equation is the solution of the differential equation that contains arbitrary constants. The general solution of an *n*th order differential equation usually contains *n* arbitrary constants.

Solving a differential equation involves the removal of the derivative terms in the equation. Since the derivative terms are results of differentiation, to solve a differential equation, the reverse process, i.e. integration is usually employed. The arbitrary constant(s) in the general solution is a consequence of integration.

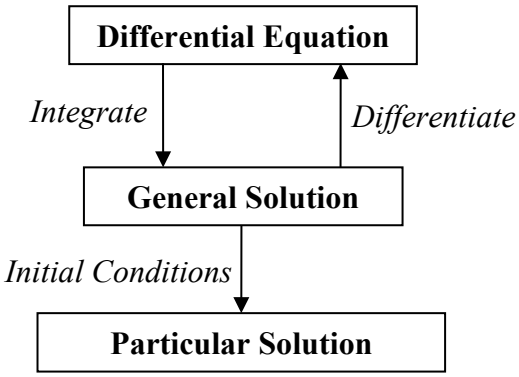


Note: The general solution of an ***n*th order** differential equation usually has ***n*** arbitrary constants.

1.3 Particular Solution

When initial conditions (i.e. the value of y corresponding to some value of x) are given, we can find a particular value of the arbitrary constant, and so obtain the **particular solution** of the differential equation.

For example, given $\frac{\mathrm{d}y}{\mathrm{d}x} = 2x$ and $y = 1$ when $x = 0$, then applying the condition given to the general solution $y = x^2 + c$, one obtains $c = 1$. Thus, one arrives at a particular solution, $y = x^2 + 1$, that solves the differential equation, *while satisfying the initial conditions*.



Hence the general procedure of solving a differential equation is as follows:

1) Determine the general solution of the differential equation by using methods illustrated in the next section: **Section 2**.
2) Apply given initial condition(s), if any, usually involving substitution of values of x, y and/or $\frac{\mathrm{d}y}{\mathrm{d}x}$, to obtain a particular solution.

2 First Order Differential Equations

2.1 Equations of the form $\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{f}(x)$

Equations of the form $\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{f}(x)$ generally can be solved by Direct Integration.

Recall that $\frac{\mathrm{d}}{\mathrm{d}x}\left[\mathrm{F}(x) + c\right] = \mathrm{f}(x) \quad \Leftrightarrow \quad \mathrm{F}(x) + c = \int \mathrm{f}(x)\ \mathrm{d}x$.

In other words, the general solution to $\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{f}(x)$ is $y = \int \mathrm{f}(x)\ \mathrm{d}x = \mathrm{F}(x) + c$ where c is an arbitrary real constant.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{f}(x) \Rightarrow y = \int \mathrm{f}(x)\ \mathrm{d}x$$

Guideline for solving $\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{f}(x)$:

Apply direct integration **once** to obtain y in terms of x. (There will be ONE arbitrary constant.)

Example 1

Find the general solutions of the differential equations **(a)** $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{x+1}$, **(b)** $\left(x^2+1\right)\frac{\mathrm{d}y}{\mathrm{d}x} = x$.

Solution:

(a) $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{x+1}$

Integrate w.r.t. x,

$$y = \int \frac{1}{x+1}\ \mathrm{d}x$$
$$= \ln\left|x+1\right| + c, \text{ where } c \in \mathbb{R}.$$

(b) $\left(x^2+1\right)\frac{\mathrm{d}y}{\mathrm{d}x} = x$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x}{x^2+1}$$

Integrate w.r.t. x,

$$y = \int \frac{x}{x^2+1}\ \mathrm{d}x$$
$$y = \frac{1}{2}\ln(x^2+1) + c, \text{ where } c \in \mathbb{R}.$$

2.2 Equations of the form $\frac{dy}{dx} = f(y)$

Equations of the form $\frac{dy}{dx} = f(y)$ generally can be solved by separating the variables.

Given $\frac{dy}{dx} = f(y)$, we divide this equation on both sides by $f(y)$, and obtain

$$\frac{1}{f(y)}\frac{dy}{dx} = 1$$

Integrating with respect to x,

$$\int \frac{1}{f(y)}\frac{dy}{dx}\,dx = \int 1\,dx$$

$$\int \frac{1}{f(y)}\,dy = \int 1\,dx$$

$$G(y) = x + c$$

The general solution is $G(y) = x + c$, where c is a real constant.

Note: Differentiating $G(y)$ with respect to y gives $\frac{1}{f(y)}$.

Alternatively, recall from **Differentiation Techniques** that

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}, \text{ so } \frac{dy}{dx} = f(y) \Leftrightarrow \frac{dx}{dy} = \frac{1}{f(y)}$$

$$x = \int \frac{1}{f(y)}\,dy$$

$$\boxed{\frac{dy}{dx} = f(y) \Rightarrow x = \int \frac{1}{f(y)}\,dy}$$

Note: For values of y where $f(y) = 0$, we need to consider constant solutions which may not be included in the above working. (This is illustrated in Example 3(ii))

Guidelines for solving $\frac{dy}{dx} = f(y)$:

1. Express the equation as $\frac{1}{f(y)}\frac{dy}{dx} = 1$ for y such that $f(y) \neq 0$.
2. Write the equation as $\int \frac{1}{f(y)}\,dy = \int 1\,dx$
3. Perform integration: L.H.S. w.r.t. y and R.H.S. w.r.t. x.
4. Obtain $x = G(y) + k$, or the form required by the question, such as, "express y in terms of x".

Example 2

Find the general solution of the differential equation $y\frac{\mathrm{d}y}{\mathrm{d}x}-1=y^2$, writing your answer in the form $y^2=\mathrm{f}(x)$. Find a particular solution given that $y=2$ when $x=0$, giving your answer in the form $y=\mathrm{g}(x)$. State also the range of values of x for which the solution is valid.

Solution:

$$y\frac{\mathrm{d}y}{\mathrm{d}x}-1=y^2$$
$$y\frac{\mathrm{d}y}{\mathrm{d}x}=y^2+1$$
$$\frac{y}{y^2+1}\frac{\mathrm{d}y}{\mathrm{d}x}=1$$

Integrating with respect to x,

$$\int \frac{y}{y^2+1}\,\mathrm{d}y=\int 1\,\mathrm{d}x$$
$$\frac{1}{2}\int \frac{2y}{y^2+1}\,\mathrm{d}y=\int 1\,\mathrm{d}x$$
$$\tfrac{1}{2}\ln\left|y^2+1\right|=x+c,\ \text{where } c\in\mathbb{R}$$
$$\ln\left(y^2+1\right)=2x+2c$$

we can remove the modulus since $y^2+1>0$

$$y^2+1=\mathrm{e}^{2x+2c}$$
$$=\mathrm{e}^{2x}\mathrm{e}^{2c}$$
$$=A\mathrm{e}^{2x},\ \text{where } A=\mathrm{e}^{2c}>0 \text{ is a constant.}$$

$\therefore$ The general solution is $y^2=-1+A\mathrm{e}^{2x}$, where A is a positive constant.

When $x=0,\ y=2: 2^2=-1+A\mathrm{e}^{2(0)}\Rightarrow A=5$

$$\therefore y^2=-1+5\mathrm{e}^{2x}$$
$$y=\pm\sqrt{5\mathrm{e}^{2x}-1}$$

Since $y=2$ when $x=0$, the particular solution is $y=\sqrt{5\mathrm{e}^{2x}-1}$.

For the solution to be valid, we require $5\mathrm{e}^{2x}-1\geq 0\Rightarrow x\geq\frac{1}{2}\ln\left(\frac{1}{5}\right)=-\frac{1}{2}\ln 5.$

Example 3

Find the general solutions, expressing y as a function of x, of the following differential equations

(i) $\dfrac{2}{2y-1}\dfrac{\mathrm{d}y}{\mathrm{d}x}=1,$ **(ii)** $2\dfrac{\mathrm{d}y}{\mathrm{d}x}=2y-1.$

Solution:

(i) $\dfrac{2}{2y-1}\dfrac{\mathrm{d}y}{\mathrm{d}x}=1$ (Note: $y\neq\frac{1}{2}$)

Integrate w.r.t x,

$$\int\frac{2}{2y+1}\,\mathrm{d}y=\int 1\,\mathrm{d}x$$

$$\ln|2y-1|=x+c,\ \text{where } c\in\mathbb{R}$$

$$|2y-1|=\mathrm{e}^{x+c}$$

$$|2y-1|=\mathrm{e}^{x}\mathrm{e}^{c}$$

$$2y-1=\pm\mathrm{e}^{c}\mathrm{e}^{x}$$

$$2y-1=A\mathrm{e}^{x},\ \text{where } A=\pm\mathrm{e}^{c},\ A\neq 0 \text{ since } y\neq\frac{1}{2}.$$

$$y=\frac{1}{2}+B\mathrm{e}^{x},\ \text{where } B=\frac{A}{2}\neq 0.$$

(ii) $2\dfrac{\mathrm{d}y}{\mathrm{d}x}=2y-1$

Case 1: If $y\neq\dfrac{1}{2}$,

$$\frac{2}{2y-1}\frac{\mathrm{d}y}{\mathrm{d}x}=1$$

From (i), we know that the general solution is

$$y=\frac{1}{2}+B\mathrm{e}^{x},\ \text{where } B=\frac{A}{2}\neq 0 \text{ is an arbitrary constant.}$$

Case 2: Consider $y=\dfrac{1}{2}$. Substituting this and its derivative (which is zero) into the differential equation $2\dfrac{\mathrm{d}y}{\mathrm{d}x}=2y-1$ gives zero on both sides, which means that $y=\dfrac{1}{2}$ is also a solution to the differential equation. This solution follows the form $y=\dfrac{1}{2}+B\mathrm{e}^{x}$, with $B=0$.

Combining Case 1 and 2, the general solution is

$$y=\frac{1}{2}+B\mathrm{e}^{x},\ \text{where } B \text{ is an arbitrary constant.}$$

2.3 Equations of the form $\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{f}(x)\mathrm{g}(y)$

A first order differential equation of the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{f}(x)\,\mathrm{g}(y), \qquad \textbf{(1)}$$

where f(x) is a function of x and g(y) is a function of y, is said to be **separable** or to have **separable variables**.

Rewriting **(1)**, we have

$$\frac{1}{\mathrm{g}(y)}\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{f}(x) \qquad \textbf{(2)}$$

provided that g(y) ≠ 0.

Integrating **(2)** with respect to x, we have

$$\int \left(\frac{1}{\mathrm{g}(y)}\frac{\mathrm{d}y}{\mathrm{d}x}\right)\mathrm{d}x = \int \mathrm{f}(x)\,\mathrm{d}x\,.$$

Hence, the general solution of the above variable separable first order differential equation is given by

$$\int \frac{1}{\mathrm{g}(y)}\mathrm{d}y = \int \mathrm{f}(x)\,\mathrm{d}x$$

which will introduce an arbitrary constant.

Remarks:

- $\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{f}(x)$ is separable since we may regard the RHS as $\mathrm{f}(x)\cdot 1$, where we consider 1 as a (very simple) function of y.

- $\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{g}(y)$ is also separable since we may regard the RHS as $\mathrm{g}(y)\cdot 1$, where we consider 1 as a (very simple) function of x.

Example 4

Find the general solution of the differential equation $y(1+x^2)\dfrac{\mathrm{d}y}{\mathrm{d}x} - 1 = y^2$.

Solution:

$$y(1+x^2)\frac{\mathrm{d}y}{\mathrm{d}x} - 1 = y^2$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \left(\frac{1}{1+x^2}\right)\left(\frac{y^2+1}{y}\right)$$

$$\frac{y}{y^2+1}\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{1+x^2}$$

Integrate w.r.t x,

$$\int \frac{y}{y^2+1}\,\mathrm{d}y = \int \frac{1}{1+x^2}\,\mathrm{d}x$$

$\dfrac{1}{2}\ln(y^2+1) = \tan^{-1}x + C$, where C is an arbitrary constant

$$\ln(y^2+1) = 2\tan^{-1}x + 2C$$

$$(y^2+1) = \mathrm{e}^{2\tan^{-1}x + 2C}$$

$$y^2 = A\mathrm{e}^{2\tan^{-1}x} - 1, \text{ where } A = \mathrm{e}^{2C}.$$

Example 5

Express y in terms of x given that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = xy - x$ and that $y = 4$ when $x = 0$.

Solution:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x(y-1)$$

$$\frac{1}{y-1}\frac{\mathrm{d}y}{\mathrm{d}x} = x$$

Integrate w.r.t x,

$$\int \frac{1}{y-1}\,\mathrm{d}y = \int x\,\mathrm{d}x$$

$\ln|y-1| = \dfrac{1}{2}x^2 + C$, where C is an arbitrary constant

$$|y-1| = \mathrm{e}^{\frac{1}{2}x^2 + C} = \mathrm{e}^{\frac{1}{2}x^2}\mathrm{e}^{C}$$

$y - 1 = A\mathrm{e}^{\frac{1}{2}x^2}$, where $A = \pm\mathrm{e}^{C}$ … (1)

Substitute $y = 4$ and $x = 0$ into equation (1) :

$$4 - 1 = A\mathrm{e}^{\frac{1}{2}(0)}$$

$$A = 3$$

Hence, $y = 3\mathrm{e}^{\frac{1}{2}x^2} + 1$.

3 Use of Substitution

Some first order differential equations which are not of the form $\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{f}(x)\mathrm{g}(y)$ can be transformed into the form by a suitable substitution. When required, the appropriate **substitution** will be **given**.

In the following Example 6, $\frac{\mathrm{d}y}{\mathrm{d}x} - y = x - 2$ is not in the form we have seen in earlier sections. By using the substitution $y = z - x$, we are able to transform the given differential equation $\frac{\mathrm{d}y}{\mathrm{d}x} - y = x - 2$ (which involves the variables y and x) to a 'new' differential equation $\frac{\mathrm{d}z}{\mathrm{d}x} = z - 1$ (which involves the variables z and x).

We solve this 'new' differential equation $\frac{\mathrm{d}z}{\mathrm{d}x} = z - 1$, which is of the form seen in Section 2.2 and obtain the general solution in terms of z and x.

To obtain the general solution for the given differential equation $\frac{\mathrm{d}y}{\mathrm{d}x} - y = x - 2$, we will just need to use the substitution again to put the general solution back in terms of the original variables y and x.

Example 6

Show that the differential equation $\frac{\mathrm{d}y}{\mathrm{d}x} - y = x - 2$ may be reduced by the means of the substitution $y = z - x$ to $\frac{\mathrm{d}z}{\mathrm{d}x} = z - 1$. Hence show that the general solution is $y = 1 - x + A\mathrm{e}^{x}$ where A is an arbitrary constant.

Solution:

$y = z - x$ ---------- (1)

Differentiate w.r.t. x,

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}z}{\mathrm{d}x} - 1$ ---------- (2)

Substitute equation (1) and (2) into $\frac{\mathrm{d}y}{\mathrm{d}x} - y = x - 2$:

$$\left(\frac{\mathrm{d}z}{\mathrm{d}x} - 1\right) - (z - x) = x - 2$$

$$\frac{\mathrm{d}z}{\mathrm{d}x} = z - 1 \text{ (shown)}$$

Case 1: If $z \neq 1$, $\displaystyle\int \frac{1}{z-1}\,\mathrm{d}z = \int 1\,\mathrm{d}x$

$$\ln|z-1| = x + c, \text{ where } c \in \mathbb{R}$$
$$|z-1| = \mathrm{e}^{x+c}$$
$$z-1 = A\mathrm{e}^{x}, \text{ where } A = \pm\mathrm{e}^{c},\ A \neq 0.$$

Case 2: $z = 1$ is also a solution to the differential equation $\dfrac{\mathrm{d}z}{\mathrm{d}x} = z - 1$ and can be written as $z - 1 = A\mathrm{e}^{x}$, where $A = 0$.

Hence the general solution is $z - 1 = A\mathrm{e}^{x}$ for any arbitrary constant A.

Since $y = z - x$, $x + y - 1 = A\mathrm{e}^{x}$.

Therefore, the general solution is $y = 1 - x + A\mathrm{e}^{x}$ where A is an arbitrary constant.

Example 7

Use the substitution $y = vx$ to find the general solution of the differential equation $\dfrac{y}{x}\dfrac{\mathrm{d}y}{\mathrm{d}x} = x + \left(\dfrac{y}{x}\right)^2$, leaving your answer in the form $y^2 = \mathrm{f}(x)$. Find the particular solution given that $y = -1$ when $x = 1$, giving your answer in the form $y = \mathrm{g}(x)$.

Solution:

$y = vx$ ---------- (1)

Differentiate w.r.t. x,

$\dfrac{\mathrm{d}y}{\mathrm{d}x} = v + x\dfrac{\mathrm{d}v}{\mathrm{d}x}$ ---------- (2)

Substitute equation (1) and (2) into $\dfrac{y}{x}\dfrac{\mathrm{d}y}{\mathrm{d}x} = x + \left(\dfrac{y}{x}\right)^2$:

$$v\left(v + x\frac{\mathrm{d}v}{\mathrm{d}x}\right) = x + v^2$$
$$v\frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

Integrate w.r.t x,

$$\int v\ \mathrm{d}v = \int 1\ \mathrm{d}x$$

$$\frac{1}{2}v^2 = x + c, \text{ where } c \in \mathbb{R}$$

$$v^2 = 2x + A, \text{ where } A = 2c$$

$$\frac{y^2}{x^2} = 2x + A$$

$$y^2 = 2x^3 + Ax^2, \text{ where } A \in \mathbb{R}$$

When $x = 1, y = -1$: $(-1)^2 = 2 + A \Rightarrow A = -1$

$$\therefore y^2 = 2x^3 - x^2$$

Since $y = -1 < 0$ when $x = 1$, the particular solution is $y = -\sqrt{2x^3 - x^2}$.

Note:

Always re-express the solution obtained by substitution in terms of the original variables *x* and *y*.

Example 8

Obtain the particular solution of the differential equation $(x + y - 1)\dfrac{\mathrm{d}y}{\mathrm{d}x} = x - y + 1$, given that $y = 4$ when $x = 0$, by using the substitution $x + y = z$.

Solution:

$x + y = z$ …… (1)

Differentiate w.r.t. *x*,

$1 + \dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\mathrm{d}z}{\mathrm{d}x}$ … … (2)

Substitute equation (1) and (2) into $(x + y - 1)\dfrac{\mathrm{d}y}{\mathrm{d}x} = x - y + 1$:

$$(z - 1)\left(\frac{\mathrm{d}z}{\mathrm{d}x} - 1\right) = x - (z - x) + 1$$

$$(z - 1)\frac{\mathrm{d}z}{\mathrm{d}x} - (z - 1) = 2x - z + 1$$

$$(z - 1)\frac{\mathrm{d}z}{\mathrm{d}x} = 2x$$

Integrate w.r.t *x*,

$$\int (z - 1)\ \mathrm{d}z = \int 2x\ \mathrm{d}x$$

$$\frac{1}{2}z^2 - z = x^2 + C, \text{ where } C \text{ is an arbitrary constant}$$

$$\frac{1}{2}(x + y)^2 - (x + y) = x^2 + C \quad \ldots\ldots (3)$$

Substitute $x=0$ and $y=4$ into (3) :

$$\frac{1}{2}(4)^2-(4)=C$$

$$C=4$$

$\therefore$ the particular solution is $\frac{1}{2}(x+y)^2-(x+y)=x^2+4$.

[OR $x^2-y^2-2xy+2x+2y=-8$]

4 Formulation and Solutions of Differential Equations

As mentioned in the introduction, several scientific and natural phenomena can be expressed as differential equations. This section discusses several applications of differential equations and focuses on the formulation of differential equations.

A key term used in most questions is "rate of change", e.g. "rate of change of volume, rate of change of flow, etc".

Example 9

Given the below scenarios, complete the following table.

Scenario	DE
The Malthusian Model suggests that the rate of change of population is proportional to the size of the population, P.	$\frac{\mathrm{d}P}{\mathrm{d}t}=kP$
Newton's Law of Cooling states that the rate of cooling of a metallic object is proportional to the difference between its temperature, T, and that of its environment, T_0 (which is a constant).	$\frac{\mathrm{d}T}{\mathrm{d}t}=-k\left(T-T_0\right)$ where k is a positive constant
The natural rate of growth (per minute) of micro-organisms in a liquid is proportional to the number of organisms, x, present. Organisms are removed at the rate of p organisms per minute.	$\frac{\mathrm{d}x}{\mathrm{d}t}=kx-p$

Example 10 (9233/2001/02/Q14b) (modified)

A rectangular tank has a horizontal base. Water is flowing into the tank at a constant rate, and flows out at a rate which is proportional to the depth of water in the tank. At time t seconds the depth of water in the tank is x metres. If the depth is 0.5 m, it remains at this constant value.

(i) Show that $\frac{\mathrm{d}x}{\mathrm{d}t} = -k(2x-1)$, where k is a positive constant.

(ii) Initially the depth of water in the tank is 0.75 m and is decreasing at a rate of 0.01 m s^{-1}. Find the time at which the depth of water is 0.55 m.

Solution:

(i) Let V m^3 denote the volume of water in the tank at any time, t.
Let A m^2 be the area of the horizontal base of the tank, which is a constant number.
Given: At time t s, the depth of water in the tank is x m.

Then $V = Ax$ where A is a positive constant.
Since rate of change of volume = rate of inflow – rate of outflow,

$$\frac{\mathrm{d}V}{\mathrm{d}t} = B - Cx \quad \text{where } B, C \text{ are positive constants.}$$

$$A\frac{\mathrm{d}x}{\mathrm{d}t} = B - Cx \quad \text{since } \frac{\mathrm{d}V}{\mathrm{d}t} = A\frac{\mathrm{d}x}{\mathrm{d}t}$$

$$\frac{\mathrm{d}x}{\mathrm{d}t} = D - Ex \quad \text{where } D = \frac{B}{A} \text{ and } E = \frac{C}{A} \text{ are positive constants.}$$

When depth (x) is 0.5 m, it remains constant, this implies that when $x = 0.5$, $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$.

Substituting $x = 0.5$, $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$, we have $D - 0.5E = 0 \Rightarrow E = 2D$

$\therefore \quad \frac{\mathrm{d}x}{\mathrm{d}t} = D - 2Dx = -D(2x-1) = -k(2x-1)$ where $D = k$. (shown) $---$ (1)

(ii) Since $x = 0.75 \neq \frac{1}{2}$, $\quad \frac{1}{2x-1}\frac{\mathrm{d}x}{\mathrm{d}t} = -k$

Integrate w.r.t t,

$$\int \frac{1}{2x-1}\,\mathrm{d}x = -\int k\,\mathrm{d}t$$

$$\frac{1}{2}\ln|2x-1| = -kt + c \text{ where } c \in \mathbb{R}$$

$$|2x-1| = \mathrm{e}^{-2kt+2c}$$

$$2x - 1 = F\mathrm{e}^{-2kt}, \text{ where } F = \pm\mathrm{e}^{2c} \in \mathbb{R} \text{ and } F \neq 0. --- (2)$$

When $t = 0,\ x = 0.75$ and $\frac{\mathrm{d}x}{\mathrm{d}t} = -0.01$, (since the depth is *decreasing*)

Substitute into equation (1): $-0.01 = -k(0.5) \Rightarrow k = 0.02$

Substitute into equation (2): $2(0.75) - 1 = F\mathrm{e}^{-0.04(0)} \Rightarrow 0.5 = F\mathrm{e}^{0} \Rightarrow F = 0.5$

$\therefore$ the particular solution is $2x-1=\frac{1}{2}\mathrm{e}^{-0.04t}$

When $x=0.55,\ 2(0.55)-1=0.5\mathrm{e}^{-0.04t}$

$$0.2=\mathrm{e}^{-0.04t}$$

$$t=\frac{\ln 0.2}{-0.04}=40.2 \text{ (3 s.f.)}$$

$\therefore$ The time is 40.2 seconds.

Example 11

At time $t = 0$, there are 8000 fishes in a lake. At time t days the birth rate of fish is equal to one-fiftieth of the number N of fishes present. Fishes are taken from the lake at the rate of 100 per day. By considering N as a continuous variable, find

(i) an expression for N at time t,

(ii) the time taken for the population of fishes in the lake to increase to 11000.

When the population of fish has reached 11000, it is decided to increase the number of fishes taken from the lake from 100 per day to F per day.

(iii) Write down the new differential equation satisfied by N.

(iv) Find the range of F such that the fishes eventually die out.

Solution:

(i) $\frac{\mathrm{d}N}{\mathrm{d}t}=\frac{1}{50}N-100=\frac{1}{50}\left(N-5000\right).$

Since $N=8000>5000,$

$$\int\frac{1}{N-5000}\mathrm{d}N=\int\frac{1}{50}\ \mathrm{d}t$$

$\ln(N-5000)=\frac{1}{50}t+c$ since $N>5000$ implies $\left|N-5000\right|=N-5000$

$$N-5000=\mathrm{e}^{\frac{t}{50}+c}$$

$N=5000+A\mathrm{e}^{\frac{t}{50}},$ where $A=\mathrm{e}^{c}>0$ is a constant

When $t=0, N=8000\Rightarrow A=3000$. Hence $N=5000+3000\mathrm{e}^{0.02t}$.

(ii) When $N = 11000$,

$$11000=5000+3000\mathrm{e}^{0.02t}$$

$$\Rightarrow 2=\mathrm{e}^{0.02t}$$

$$\Rightarrow t=50\ln 2=34.7 \text{ (3 s.f)}$$

Hence it takes 34.7 days for the population of fishes in the lake to increase to 11000.

(iii) $\dfrac{\mathrm{d}N}{\mathrm{d}t} = \dfrac{1}{50}N - F.$

(iv) We need $\dfrac{\mathrm{d}N}{\mathrm{d}t} < 0$ for the fishes to eventually die out. This means we need

$\dfrac{\mathrm{d}N}{\mathrm{d}t} = \dfrac{1}{50}(11000) - F < 0 \Rightarrow F > 220.$ This will ensure that $\dfrac{\mathrm{d}N}{\mathrm{d}t} < 0$ when $N = 11000$, and since N decreases, $\dfrac{\mathrm{d}N}{\mathrm{d}t} = \dfrac{1}{50}N - F < 0$ until the fishes eventually die out.

Example 12

Suppose that a chemical mixture contains two substances whose masses are x kg and y kg, and whose combined mass is always 1 kg. At any time t, the rate at which x is increasing is proportional to the product of the two masses at that time.

(i) Obtain a differential equation relating x and t.

(ii) Show that the general solution of the differential equation may be expressed in the form $\ln\left(\dfrac{Cx}{1-x}\right) = kt$, where C and k are positive constants.

(iii) Given that initially $x = \dfrac{1}{10}$ and when $t = 1$, $x = \dfrac{1}{4}$, express x in terms of t.

(iv) State what happens to x for large values of t. Sketch a graph of x against t.

Solution:

(i) $x + y = 1 \Rightarrow y = 1 - x$

$\dfrac{\mathrm{d}x}{\mathrm{d}t} = kxy = kx(1-x)$ for some positive constant k.

(ii) $$\int \frac{1}{x(1-x)}\,\mathrm{d}x = k\int 1\,\mathrm{d}t$$

Using partial fractions, $\dfrac{1}{x(1-x)} = \dfrac{1}{x} + \dfrac{1}{1-x}$,

$$\int\left(\frac{1}{x} + \frac{1}{1-x}\right)\mathrm{d}x = k\int 1\,\mathrm{d}t$$

$\ln|x| - \ln|1-x| + \ln C = kt$ where $C > 0$ (we need $C > 0$ for $\ln C$ to be defined)

$$\ln\left|\frac{Cx}{1-x}\right| = kt$$

$\ln\left(\dfrac{Cx}{1-x}\right) = kt$, since $0 < x < 1$, $\dfrac{Cx}{1-x} > 0$ $\quad$ (shown)

(iii) When $t = 0,\ x = \frac{1}{10}$: $\ln\frac{C}{9} = 0 \Rightarrow C = 9$

When $t = 1,\ x = \frac{1}{4}$: $\ln\frac{C}{3} = k \Rightarrow k = \ln 3$

$$\ln\frac{9x}{1-x} = t\ln 3 \Rightarrow \ln\frac{9x}{1-x} = \ln 3^t$$

$$\Rightarrow \frac{9x}{1-x} = 3^t \Rightarrow 9x = 3^t(1-x)$$

$$\Rightarrow x = \frac{3^t}{9+3^t}$$

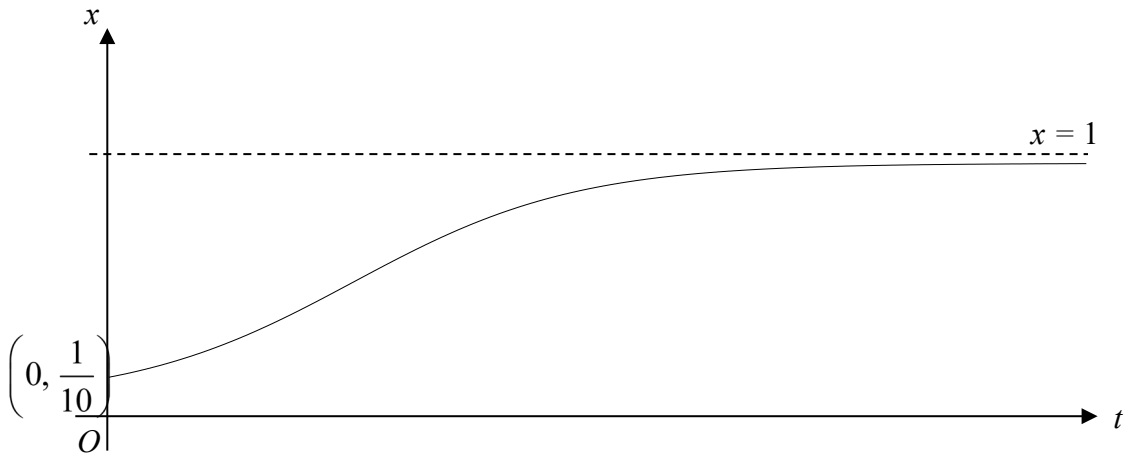
(iv) $x = \frac{3^t}{9+3^t}$

Divide both numerator and denominator by 3^t

$$= \frac{1}{\frac{9}{3^t}+1} = \frac{1}{3^{2-t}+1}$$

As $t \to \infty,\ 3^{2-t} \to 0 \Rightarrow x \to \frac{1}{0+1} = 1$.

For large values of t, x approaches the value of 1.



Note: In the context of the question, we only sketch the graph for values of $t \geq 0$.

Application to Problems that involve Motion of a Particle

Equations of the form $\frac{\mathrm{d}x}{\mathrm{d}t} = \mathrm{f}(t)$ and $\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = \mathrm{f}(t)$ arise in the study of the motion of a particle in a straight line. The directed distance from a fixed point O is called its **displacement,** x.

The rate at which the displacement increases with respect to time, $\frac{\mathrm{d}x}{\mathrm{d}t}$, is represented by v and is called the **velocity** of the particle, while the **acceleration** $a = \frac{\mathrm{d}v}{\mathrm{d}t}$ is the rate at which the velocity increases with respect to time, *t*.

Example 13

A particle moves along the *x*-axis. It passes the origin O with a velocity of 8 units/s and its acceleration *t* seconds after passing O is given by $a = (3-t)$ units/s^2.

Calculate

(i) the velocity of the particle when $t = 2$,

(ii) the maximum velocity of the particle,

(iii) the displacement of the particle from O when it is instantaneously at rest.

Solution:

(i) Since $a = \frac{\mathrm{d}v}{\mathrm{d}t}$,

Integrate w.r.t *t*,

$$v = \int a \, \mathrm{d}t$$
$$= \int (3-t) \, \mathrm{d}t$$
$$= 3t - \frac{t^2}{2} + A, \text{ where } A \in \mathbb{R}.$$

When $t = 0$, $v = A = 8$. $\therefore$ $v = -\frac{t^2}{2} + 3t + 8$

When $t = 2$, $v = -\frac{(2)^2}{2} + 3(2) + 8 = 12$ units/s

(ii) For maximum velocity, $\frac{\mathrm{d}v}{\mathrm{d}t} = 0 \Rightarrow 3 - t = 0 \Rightarrow t = 3$

When $t = 3$, $v = -\frac{(3)^2}{2} + 3(3) + 8 = \frac{25}{2}$ units/s

$\frac{\mathrm{d}^2v}{\mathrm{d}t^2} = -1 < 0$ so velocity is indeed a maximum.

(iii) $v = \dfrac{\mathrm{d}x}{\mathrm{d}t}$, where x is displacement of the particle from O.

$$x = \int v\,\mathrm{d}t = \int \left(-\frac{t^2}{2} + 3t + 8 \right) \mathrm{d}t$$

$$= -\frac{t^3}{6} + \frac{3t^2}{2} + 8t + B, \text{ where } B \in \mathbb{R}$$

When $t = 0, x = 0 = B$. $\therefore x = -\dfrac{t^3}{6} + \dfrac{3t^2}{2} + 8t$.

The particle is at rest when $v = 0$, i.e. $-\dfrac{t^2}{2} + 3t + 8 = 0$

$$-\frac{1}{2}(t-8)(t+2) = 0$$

Since $t > 0,\ t = 8$.

$$\therefore x = -\frac{(8)^3}{6} + \frac{3(8)^2}{2} + 8(8) = 74\frac{2}{3} \text{ units}$$

CONCLUSION

This is just a short introduction to differential equations, and the world of mathematical modelling and numerical simulation takes on considerable importance in the last few decades in all domains of science and industrial applications (or engineering sciences).

Mathematical modelling is the art (or science, depending on point of view) of representing (or transforming) a physical reality into abstract models which are accessible to analysis and calculation. However, the reality is many of these models cannot be solved easily by hand, and thus numerical simulation is especially important; a process which allows calculating on the computer the solutions of these models, and thus to simulate the physical reality.

In fact, everything changed with the invention of computers after the second world war. The computer has been a revolution for mathematics more than other disciplines: it has effectively opened a new field, that of modelling and simulation. The computer has made it possible for mathematics to be an experimental science (where we carry out numerical experiments like what other sciences do in physical experiments), and the conception as well as the analysis of the methods of calculation on the computer have became a new branch of mathematics : numerical simulation.

These progress have also allowed mathematics to attack problems which are much more complex and concrete, directly motivated from the industry or science, to which we can bring about answers which are both qualitative and quantitative; that is mathematical modelling.

SUMMARY